

BANK LOAN DEFAULT RISK ANALYSIS

Data Dictionary

Complete Reference for All 33 Dataset Columns + 7 Engineered Features

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This data dictionary documents all 33 columns in the raw Bank Loan Default Risk dataset. Each entry includes the column name, data type, business definition, sample values, and analytical use. Seven additional engineered features created during data preparation are listed in Section 7.

5,100	5,000	100	33	40	7
Raw Records	Cleaned Records	Duplicates Removed	Raw Columns	Final Columns	Engineered Features

1. Customer Demographics

Column Name	Data Type	Business Definition	Sample Values	Analytical Use
Customer_ID	Text	Unique identifier for each loan applicant. Primary key linking all records.	CUST_00001, CUST_99999	Deduplication; join key across tables
Age	Integer	Borrower's age in years at time of loan application.	22, 35, 58	Age-band segmentation; demographic risk analysis
Gender	Text	Borrower's gender identity.	Male, Female	Gender-based default rate comparison
Marital_Status	Text	Borrower's marital status at time of application.	Married, Single, Divorced	Household financial stability indicator
Dependents	Integer	Number of financially dependent individuals (children, elderly) supported by the borrower.	0, 1, 2, 3, 4	Disposable income estimation; affordability analysis
Education	Text	Highest educational qualification attained by the borrower.	Graduate, Post Graduate, High School, Unknown	Income potential proxy; credit risk segmentation

Column Name	Data Type	Business Definition	Sample Values	Analytical Use
City_Tier	Text	Classification of borrower's city by economic development level.	Tier 1, Tier 2, Tier 3	Geographic risk segmentation; regional default analysis

2. Employment Information

Column Name	Data Type	Business Definition	Sample Values	Analytical Use
Occupation	Text	Borrower's stated profession or job role.	Salaried, Self Employed, Student, Business	Occupational income stability indicator
Employment_Status	Text	Current employment classification of the borrower.	Employed, Self Employed, UnEmployed, Unknown	Key risk driver; income reliability assessment
Years_of_Employment	Integer	Total number of years the borrower has been continuously employed in their current role.	0, 1, 5, 12, 39	Job stability measure; creditworthiness indicator

3. Financial Profile

Column Name	Data Type	Business Definition	Sample Values	Analytical Use
Annual_Income	Float	Total gross annual income of the borrower in Indian Rupees.	1,50,000, 4,80,000, 12,00,000	Repayment capacity; eligibility assessment
Monthly_Income	Float	Monthly gross income derived from annual income (Annual Income / 12).	12,500, 40,000, 1,00,000	EMI affordability check; DTI calculation
Monthly_Expenses	Float	Average monthly expenditure reported by the borrower.	8,000, 22,000, 65,000	Disposable income calculation; financial stress indicator
Savings_Balance	Float	Total liquid savings held by the borrower at time of application, in Rupees.	5,000, 75,000, 3,50,000	Liquidity buffer; emergency repayment capacity
Existing_Loan_Amount	Float	Total outstanding principal across all existing loans held by the borrower.	0, 50,000, 5,00,000	Debt burden; total exposure assessment
Existing_EMI	Float	Monthly repayment obligation on all currently active loans.	0, 3,500, 18,000	DTI ratio input; affordability analysis

Column Name	Data Type	Business Definition	Sample Values	Analytical Use
Credit_Card_Balance	Float	Outstanding unpaid balance across all credit cards held by the borrower.	0, 12,000, 85,000	Revolving debt indicator; credit utilisation analysis
Debt_to_Income_Ratio	Float	Percentage of monthly income consumed by all debt repayments. Formula: (Total EMI / Monthly Income) x 100.	18.5, 35.0, 62.3	Primary financial risk metric; default predictor

4. Credit History

Column Name	Data Type	Business Definition	Sample Values	Analytical Use
Credit_Score	Integer	Numerical creditworthiness score based on repayment history and debt levels. Range: 300 to 900.	300, 480, 602, 780, 899	Strongest single predictor of default risk
Credit_History_Years	Float	Number of years the borrower has maintained an active credit profile.	0.5, 5.0, 20.0	Credit maturity indicator; risk profiling
Missed_Payments_Last_12M	Integer	Number of scheduled loan or credit card payments missed in the past 12 months.	0, 1, 3, 7	Recent payment behaviour; delinquency risk indicator
Previous_Defaults	Integer	Total number of times the borrower has defaulted on any previous loan obligation.	0, 1, 2	Historical default flag; critical risk predictor
Number_of_Credit_Cards	Integer	Total count of credit cards currently held by the borrower.	0, 2, 5, 8	Credit utilisation breadth; financial exposure indicator

5. Loan Details

Column Name	Data Type	Business Definition	Sample Values	Analytical Use
Loan_ID	Text	Unique identifier for each individual loan application.	LN_000001, LN_005000	Loan-level tracking; deduplication
Loan_Type	Text	Category of loan product applied for by the borrower.	Auto, Home, Personal, Education	Loan-product risk comparison; portfolio segmentation
Loan_Amount	Float	Total principal amount requested by the borrower in Indian Rupees.	50,000, 5,00,000, 25,00,000	Exposure analysis; high-value loan risk assessment

Column Name	Data Type	Business Definition	Sample Values	Analytical Use
Loan_Term_Months	Integer	Agreed repayment duration of the loan in months.	12, 24, 36, 60, 120, 240	Repayment timeline; long-term risk assessment
Interest_Rate	Float	Annual interest rate applied to the loan, expressed as a percentage.	7.5, 10.2, 14.8	Pricing analysis; rate-to-default correlation
EMI	Float	Fixed monthly repayment amount due from the borrower for the current loan.	3,200, 12,500, 45,000	Repayment burden; affordability check

6. Loan Outcome and Risk Metrics

Column Name	Data Type	Business Definition	Sample Values	Analytical Use
Loan_Status	Text	Final decision outcome of the loan application, indicating whether the application was approved or rejected by the lending institution.	Approved, Rejected	Loan approval analysis; application outcome tracking
Defaulted	Text	Binary indicator of whether the borrower has defaulted on the current loan.	Yes, No	Primary target variable for default analysis
Days_Past_Due	Integer	Number of calendar days the borrower is overdue on the most recent payment.	0, 15, 90, 180	Delinquency severity; early warning signal
Risk_Category	Text	Risk classification assigned to each loan based on credit and financial profile.	High, Low	Executive reporting; portfolio risk segmentation

7. Engineered Features — Created During Data Preparation

Column Name	Data Type	Business Definition	Sample Values	Analytical Use
Defaulted_Flag	Integer	Binary numeric encoding of the Defaulted column. Yes = 1, No = 0.	0, 1	KPI calculations; dashboard default rate metric
Credit_Score_Band	Text	Categorical segmentation of credit scores into risk quality tiers.	Poor, Average, Good, Fair, Excellent	Credit risk profiling; segment-level default analysis
DTI_Band	Text	Debt-to-Income ratio grouped into debt burden ranges.	0-20%, 21-40%, 41-60%, 61%+	Financial stress segmentation; DTI risk dashboard

Column Name	Data Type	Business Definition	Sample Values	Analytical Use
Loan_Amount_Band	Text	Loan amount grouped into exposure size brackets, in Rupees.	0-2 Lakh, 2-5 Lakh, 5-10 Lakh, 10-20 Lakh, 20 Lakh+	Exposure analysis; loan size risk comparison
Interest_Rate_Band	Text	Interest rate grouped into pricing tier ranges.	Below 8%, 8-10%, 10-12%, 12-14%, Above 14%	Rate-tier default comparison; pricing risk analysis
EMI_Band	Text	Monthly EMI grouped into repayment burden brackets, in Rupees.	0-5,000, 5,001-10,000, 10,001-20,000, 20,001-40,000, Above 40,000	Repayment burden segmentation
Savings_Balance_Band	Text	Savings balance grouped into liquidity strength tiers, in Rupees.	0-25,000, 25,001-50,000, 50,001-1 Lakh, 1-5 Lakh, 5 Lakh+	Liquidity risk profiling; savings-to-default analysis